

Record-Efficiency Inverted CsPbI₃ Perovskite Solar Cells Enabled by Rearrangement and Hydrophilic Modification of SAMs

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Recently, the inverted CsPbI₃ perovskite solar cells (PSCs) have attracted extensive attentions due to their potential to combine with silicon solar cells for tandem devices with theoretical power conversion efficiency (PCE) of 44%. However, the reported self-assembled molecules (SAMs) as hole selected layer for inverted PSCs have poor wettability and serious agglomeration, which greatly limits the efficiency and stability of PSCs. To address above problem, niobium pentachloride (NCL) is applied to prevent SAMs agglomeration for a homogenous SAMs film with a hydrophilic surface. The optimized SAMs surface facilitates the deposition of the cesium lead triiodide (CsPbI₃) film with an enhanced referred orientation, suppressed defects, and released stress. Consequently, the NCL-treated CsPbI₃ PSCs achieved a champion PCE of 21.24%, which is the highest PCE value for inverted all-inorganic PSCs. The optimized device maintained 97.61% initial PCE after 1000 h storage in air, and 92.27% initial PCE after 1000 h tracking at the maximum power point (MPP).

based on cesium lead triiodide (CsPbI₃) stand out due to their suitable bandgap (≈ 1.70 eV) for tandem solar cell applications and excellent long-term stability originating from the elimination of volatile organic compounds and unstable Spiro-OMeTAD present in traditional n-i-p hybrid PSCs.^[2] The hole transport layers (HTLs) of inverted PSCs typically require the use of self-assembled molecules (SAMs) such as [2-(9H-carbazol-9-yl)ethyl]phosphonic acid (2PACz), and [2-(3,6-dimethoxy-9H-carbazol-9-yl)ethyl]phosphonic acid (MeO-2PACz).^[3] However, SAMs, as amphiphilic materials, tend to accumulate on substrates, leading to inhomogeneous distribution and rough surface morphology, resulting in poor-quality perovskite films and a serious open-circuit voltage (V_{oc}) loss in the PSCs.^[4] Additionally, the unfavorable buried interface contact and inappropriate energy levels pose challenges to enhancing the performance of PSCs.^[5]

1. Introduction

Perovskite solar cells (PSCs), as one of the most promising technologies in the field of photovoltaics, have witnessed significant advancements over the years, with a dramatic increase in power conversion efficiency (PCE) from 3.8% in 2009 to 26.54% in 2024.^[1] Among these developments, the inverted PSCs

To address the aforementioned issues, Guo et al. proposed a plasma treatment strategy to regulate the surface morphologies of substrates for achieving conformal growth of SAMs and effectively inhibiting non-radiative recombination at the interface.^[6] Luo et al. had the aggregation of SAMs alleviated by simply treating the SAMs layer with a solution of formamidinium chloride (FACl) dissolved in N,N-dimethylformamide (DMF) solvent.^[7] Liu et al. have developed a co-solvent strategy to disassemble the micelles of SAMs solution for optimizing the quality of SAMs film.^[8] Tang et al. synthesized a novel SAM with a trimethoxysilane group that exhibited a strong tridentate anchoring to the substrate. The new SAM based PSCs exhibited a competitive PCE and respectable stability.^[9] In addition to improving the performance of inverted PSCs through substrate modification or regulating SAMs dispersion as described above, the effective strategies for defect passivation and lattice stress release at the buried interface are also important to improve device performance.^[10] As a note, at present, it is still a blank in the research on the preparation of efficient inverted inorganic PSCs through modified SAMs and optimized buried interface.

In this work, niobium pentachloride (NCL) was introduced to enhance the quality of the SAMs by facilitating a homogenous distribution, thus improving the performance of the inverted

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PSCs. Atomic force microscope (AFM) and sum frequency generation (SFG) vibrational spectroscopy analyses revealed the interaction of NCL with SAMs, which helps prevent SAMs clustering, aids in uniform distribution, and enhances interfacial contact. Furthermore, the grazing-incidence wide-angle X-ray scattering (GIWAXS) measurement confirmed that NCL promotes the deposition of the perovskite precursor, facilitating the optimal growth of CsPbI₃ films along the [110] direction. Ultimately, the NCL-treated PSCs exhibited improved PCE and stability, achieving a champion PCE of 21.24%, which is the highest PCE reported for all-inorganic inverted PSCs, with the V_{OC} as high as 1.25 V and the energy loss (E_{loss}) as low as 0.48 eV. In terms of the ambient stability and operational stability, the NCL-treated devices maintained 97.61% of the initial PCE after 1000 h storage, and 92.27% of the initial PCE after tracking for 1000 h at the maximum power point (MPP).

2. Results and Discussion

The niobium pentachloride (NCL) was spin-coated onto the prepared self-assembled molecules (SAMs) substrates to regulate their surface morphology and thus achieve more uniform distributions. The agglomerated SAMs clusters are attributed to the amphiphilic nature of SAMs which originates from their polar phosphate (PA) groups and nonpolar carbazole groups.^[8] When SAMs were dissolved in isopropanol (IPA) solvent, their PA groups face outward toward the IPA molecules while their nonpolar carbazole groups face inward.^[11] After spin-coating SAMs solution on the Glass/Indium-Tin Oxide (ITO)/Nickel Oxide (NiO_x) substrate and annealing to evaporate the IPA, SAMs remain on Glass/ITO/NiO_x substrate as clusters of globular micelles. In this work, the NCL in isopropanol (IPA) solution was spin-coated on Glass/ITO/NiO_x/SAMs substrate, which could interact with the SAMs by forming a chelated structure, leading to an rearrangement of SAMs for an uniform distribution of SAMs over the entire substrate. The effect of NCL on SAMs dispersion was investigated by spin-coating NCL solution (0.50–2.00 mg mL⁻¹ in IPA) onto Glass/ITO/NiO_x/SAMs substrates. The top-view scanning electron microscopy (SEM) images of Glass/ITO/NiO_x/SAMs substrates with different concentrations of NCL modification are illustrated in Figure 1a. The morphology of Glass/ITO/NiO_x/SAMs films was significantly changed after NCL treatment. The NCL-modified substrates exhibited a more uniform surface coverage than the control substrate without NCL modification, and this effect was more pronounced with increasing NCL concentration. To account for the solvent's effect, SEM images were also captured for the substrates with and without IPA treatment (Figure S1, Supporting Information). As comparison, the spreading function of NCL solution treatment (Figure 1a) is much more obvious than only IPA treatment.

To further illustrate the spreading effect of NCL on SAMs distribution, energy dispersive X-ray spectroscopy (EDS) mapping was conducted to probe the distribution of SAMs. As shown in Figure S2 (Supporting Information), the distributions of Cl and Nb were detected on the surface of the NCL-treated Glass/ITO/NiO_x/SAMs substrate, confirming the existence of NCL. Compared with the control substrate without treatment, a higher percentage of P and a lower percentage of Ni indicate that SAMs were more uniformly distributed on the treated sub-

strate and covered the exposed NiO_x after NCL treatment. The insets of Figure 1a show the contact angles of the substrates with different concentrations of NCL treatment. The contact angle decreases with increasing NCL concentration, the more hydrophilic surface is more beneficial for the deposition of perovskite film.^[12] Figure 1b presents the atomic force microscopy (AFM) images of the Glass/ITO/NiO_x/SAMs substrates with NCL treatment. The roughness also decreases with increasing NCL concentration, illustrating a reduced agglomeration of SAMs.

To further evaluate the effect of NCL-treated substrate on top perovskite films, as shown in Figure 1c,d, SEM and AFM images of cesium lead triiodide (CsPbI₃) films were obtained. Compared with the control perovskite film, the perovskite films on NCL-treated substrates have denser, flatter, and more hydrophilic surfaces with fewer pinholes, which is conducive to the stability improvement of CsPbI₃ films in ambient air.^[13]

After revealing the effect of NCL on the morphology of Glass/ITO/NiO_x/SAMs substrates and perovskite films, the effect of NCL on surface potentials of Glass/ITO/NiO_x/SAMs substrates was measured using Kelvin probe force microscopy (KPFM). As shown in Figure 2a,b, the contact potential difference (CPD) of Glass/ITO/NiO_x/SAMs was elevated after NCL treatment, implying a lower work function for the NCL-treated Glass/ITO/NiO_x/SAMs substrate.^[14] This can be attributed to the fact that NCL allows for a flatter SAMs spread and an optimized SAMs film quality.^[15]

In order to estimate the effect of the NCL treatment on the crystallinity of perovskite films, ultraviolet-visible absorption spectra (UV-vis) and X-ray diffraction (XRD) measurements were tested. As shown in Figure 2c–e, NCL treatment did not change the crystal bandgap and structure of CsPbI₃ film, but enhanced the absorption and the diffraction peaks of (110) and (220) planes of the perovskite film. The 1.00 mg mL⁻¹ NCL treated perovskite film (target film) exhibited the highest absorption and diffraction intensity. The grazing-incidence wide-angle X-ray scattering (GIWAXS) was conducted to confirm the above results. As shown in Figure 2f,g, the Debye-Scherrer rings located near $q_z = 0.9 \text{ \AA}^{-1}$ and $q_z = 1.8 \text{ \AA}^{-1}$ correspond to the (110) and (220) crystal planes of perovskite film, respectively. The control film displays uniform intensity across the diffraction ring at $q_z = 0.9 \text{ \AA}^{-1}$, while the target film shows sharp and discrete Bragg spots on the diffraction ring, indicating that NCL treatment is beneficial for promoting the optimal growth of perovskite films along the [110] direction, which is consistent with XRD results.^[16]

Then the effect of NCL treatment on the residual stress (σ_M) of the perovskite film was investigated using the grazing incidence X-ray diffraction (GIXRD) technique. As shown in Figure S3 (Supporting Information), the diffraction peaks at $2\theta = 28.8^\circ$ for the control film gradually shifted toward smaller angles as the incident angles increased from 0.5° to 1.4° , indicating the existence of serious tensile stress in the control film. In contrast, the shift degree was noticeably reduced for the target film, which could be attributed to the improved interfacial contact.^[17]

To illustrate the impact of NCL treatment on photo-generated charge transfer at the buried interface of the perovskite film, steady state photoluminescence (PL) and time-resolved photoluminescence (TRPL) measurements were conducted. As shown in Figure 2h, the intensity of PL peak was significantly reduced after NCL treatment, and the target film exhibits the lowest

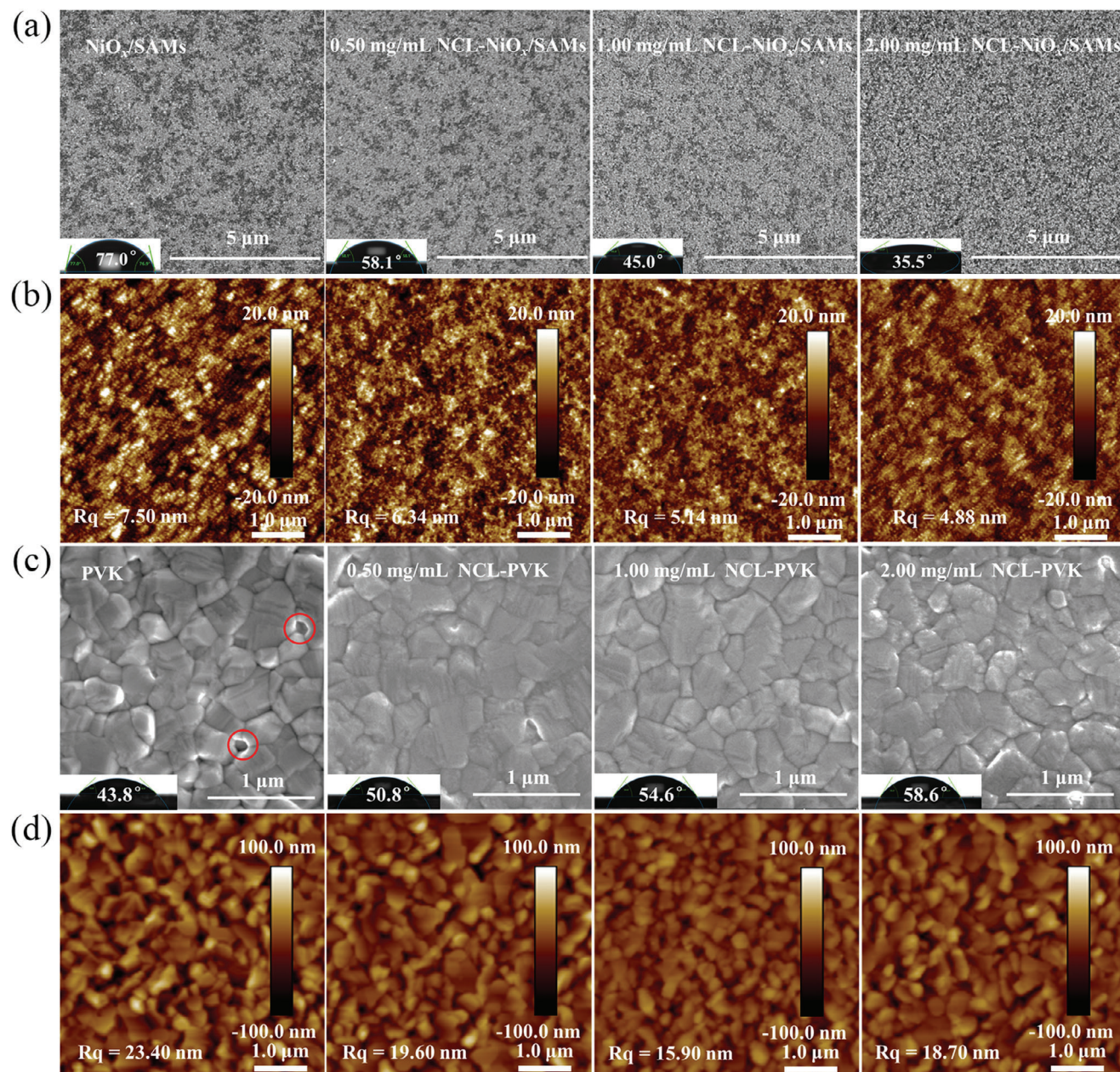


Figure 1. a) Top-view SEM images and contact angles and b) AFM images of Glass/ITO/NiO_x/SAMs films with different concentrations of NCL treatment. c) Top-view SEM images and contact angles and d) AFM images of Glass/ITO/NiO_x/SAMs/PVK films with different concentrations of NCL treatment.

intensity, indicating that NCL treatment is favorable for the hole transfer from the perovskite layer to the hole transport layer (HTL). TRPL curves are fitted using a double-exponential decay function (Figure 2i; Table S1, Supporting Information). In line with the PL results, the target film shows the shortest carrier lifetime. In addition, the PL and TRPL spectra of Glass/NCL/CsPbI₃ are also tested (Figure S4 and Table S2, Supporting Information). The perovskite films on NCL-treated glass substrates show enhanced PL intensity and prolonged carrier lifetime, indicating that NCL has a passivation effect on the defects of perovskite.

To elucidate the reason of the homogeneous dispersion of SAMs after NCL treatment, the sum frequency generation (SFG) vibrational spectroscopy, a cutting-edge nonlinear interface-sensitive spectroscopy, was conducted to detect the vibration modes of various functional groups in SAMs.^[18] SFG can selectively and nondestructively probe the buried solid/solid interface in situ by using a reflection sample geometry. This approach allows for the exploration of the interfacial structure of Glass/NiO_x/SAMs at a molecular level, and enables the visual analysis of the orientation of molecules or functional groups at the interface.^[19] As shown in Figure 3a, the vibration peak of

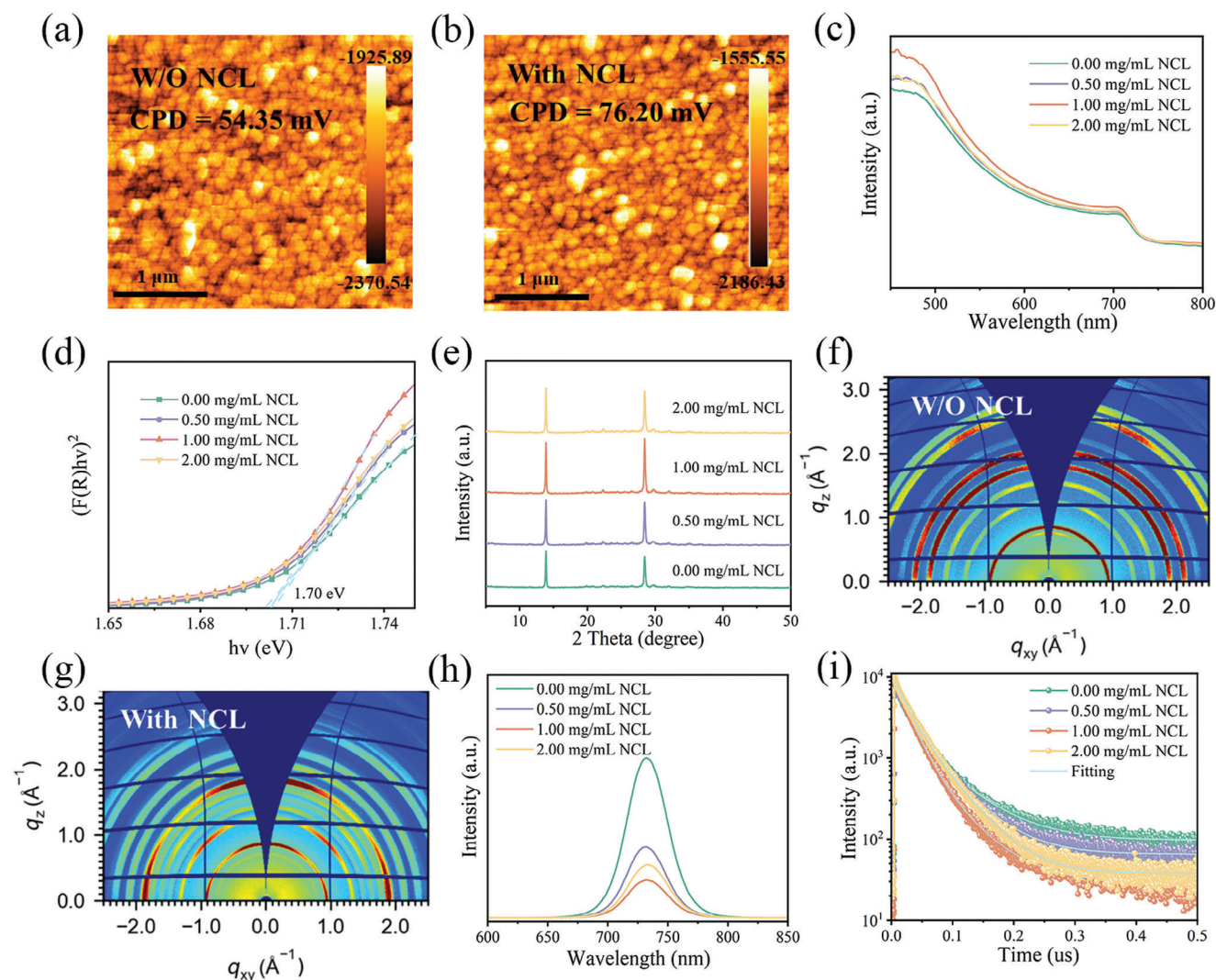


Figure 2. KPFM images for a) control and b) NCL-treated Glass/ITO/NiO_x/SAMs films. c) UV-visible absorption spectra, d) Tauc spectra for Glass/ITO/NiO_x/SAMs/PVK films with different concentrations of NCL treatment. e) XRD patterns of Glass/ITO/NiO_x/SAMs/PVK films with different concentrations of NCL treatment. GIWAXS images of f) control and g) NCL-treated Glass/ITO/NiO_x/SAMs/PVK films. h) The steady-state PL spectra and i) TRPL curves for Glass/NiO_x/SAMs/PVK films with different concentrations of NCL treatment.

C–H group at $\approx 2945\text{ cm}^{-1}$ was detected in the pure SAMs film, which was intensified after NCL treatment, and attributed to the improved orientation of SAMs molecules.^[20] In addition, the C–H signal was shifted after NCL treatment, which was consistent with the result of Fourier-transform infrared (FTIR) in Figure S5 (Supporting Information). Furthermore, Figure S6 (Supporting Information) demonstrates the impact of NCL concentrations on the SFG signal of SAMs. With increasing NCL concentration, the SFG signal of SAMs gradually increases, peaking at 1.00 mg mL^{-1} NCL treatment.

In order to further characterize the interaction between the SAMs and NCL, FTIR was conducted. Given that the SAMs consist of [2-(9H-carbazol-9-yl)ethyl]phosphonic acid (2PACz) and [2-(3,6-Dimethoxy-9H-carbazol-9-yl)ethyl]phosphonic acid (MeO-2PACz), the interaction of NCL with 2PACz or MeO-2PACz was examined separately. As shown in Figure S7 (Supporting Information) and Figure 3b,c, the vibration peaks of the P–OH stretch-

ing vibration for 2PACz and MeO-2PACz are detected at 1007.78 and 1019.35 cm^{-1} , respectively. Both of them shifted to lower wavenumber due to interactions with NCL.^[3b,21] Meanwhile, the vibration peak corresponding to P=O also shifted (Figure S8, Supporting Information). The shifts of P–OH and P=O peaks reflect strong interactions between SAMs and NCL.^[12a,22]

X-ray photoelectron spectra (XPS) provide additional insights into the interaction between SAMs and NCL. The high-resolution XPS spectra of P 2p for 2PACz and MeO-2PACz are shown in Figure 3d,e. The two P 2p peaks shift toward lower binding energies and exhibit a significant decrease in intensity after NCL modification, confirming the strong interaction between NCL and SAMs. As the Nb 3d spectra of NCL shown in Figure 3f, two peaks at 209.92 and 207.19 eV are assigned to Nb $3d_{3/2}$ and Nb $3d_{5/2}$, respectively,^[23] which shift to higher binding energies after interaction with SAMs. Furthermore, the two characteristic peaks of Cl 2p for NCL shift to higher binding energies (Figure 3g).

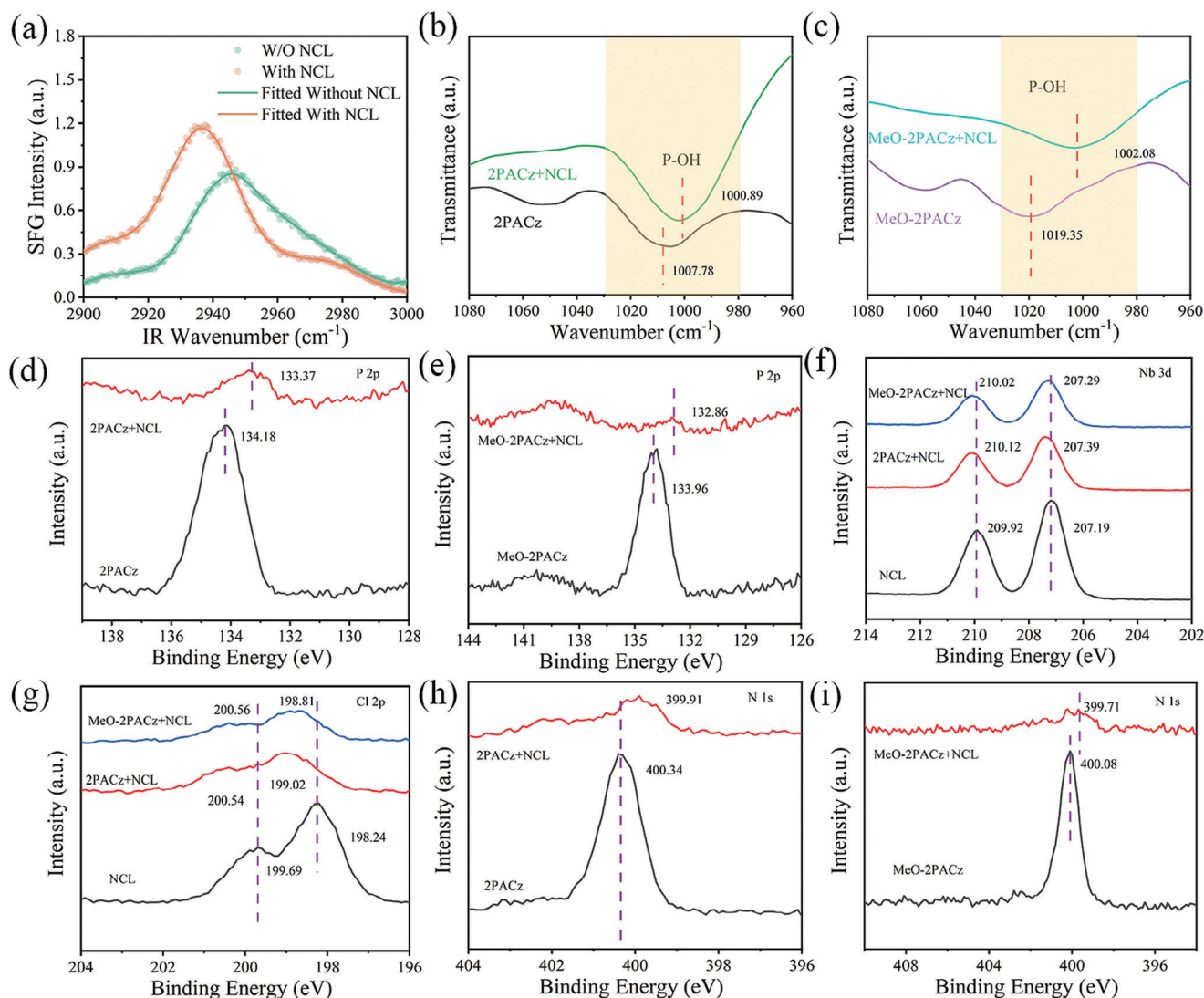


Figure 3. a) SFG spectra for the control and NCL-treated Glass/ NiO_x /SAMs films. The enlarged region of FTIR spectra for b) 2PACz and 2PACz + NCL films, c) MeO-2PACz and MeO-2PACz + NCL films. XPS spectra of P 2P for d) 2PACz and 2PACz + NCL films, e) MeO-2PACz and MeO-2PACz + NCL films. XPS spectra of f) Nb 3d and g) Cl 2p for NCL, 2PACz + NCL, and MeO-2PACz + NCL films. XPS spectra of N 1s for h) 2PACz and 2PACz + NCL films, i) MeO-2PACz and MeO-2PACz + NCL films.

The shifts of Nb and Cl further confirm the strong interaction between NCL and SAMs. The interactions between NCL and carbazole in SAMs are characterized by the N 1s spectra of 2PACz and MeO-2PACz.^[24] As depicted in Figure 3h,i, after mixing NCL with SAMs, the two N 1s peaks of 2PACz and MeO-2PACz shift to lower binding energies by 0.43 and 0.37 eV, respectively.

To demonstrate the photovoltaic performance of the NCL-treated devices, inverted perovskite solar cells (PSCs) were fabricated with the configuration of ITO/ NiO_x /SAMs/NCL/ CsPbI_3 /[6,6]-phenyl-C61-butyric acid methyl ester (PC_{61}BM)/bathocuproine (BCP)/Ag (Figure 4a). The ultraviolet photoelectron spectroscopy (UPS) was conducted to investigate the band structure of the fabricated inverted PSCs. As depicted in Figures 4b and S9 and Table S3 (Supporting Information), the UPS results reveal that the valence band (V_B) of the HTL shifts from -5.44 to -5.47 eV after NCL treatment,

bringing it closer to the V_B of CsPbI_3 (-5.52 eV), which is conducive to a higher open-circuit voltage (V_{OC}) for CsPbI_3 PSCs. Figures 4c and S10 (Supporting Information) display the $J-V$ curves of inverted PSCs with and without NCL treatment. With the increase of NCL concentration, the efficiency of the inverted PSCs gradually improves, reaching the highest power conversion efficiency (PCE) of 21.24% at the concentration of 1.00 mg mL^{-1} , with an open circuit voltage (V_{OC}) of 1.25 V, a fill factor (FF) of 83.10% and a short circuit current (J_{SC}) of 20.50 mA cm^{-2} (Table S4, Supporting Information). As shown in Table S5 (Supporting Information), the target device exhibits an energy loss (E_{loss}) as low as 0.48 eV, marking the highest reported PCE for all-inorganic inverted PSCs. The enhanced performance of device with NCL treatment can be attributed to better energy level matching and improved hole extraction. The integrated current densities (J_{int}) of the inverted PSCs without and with

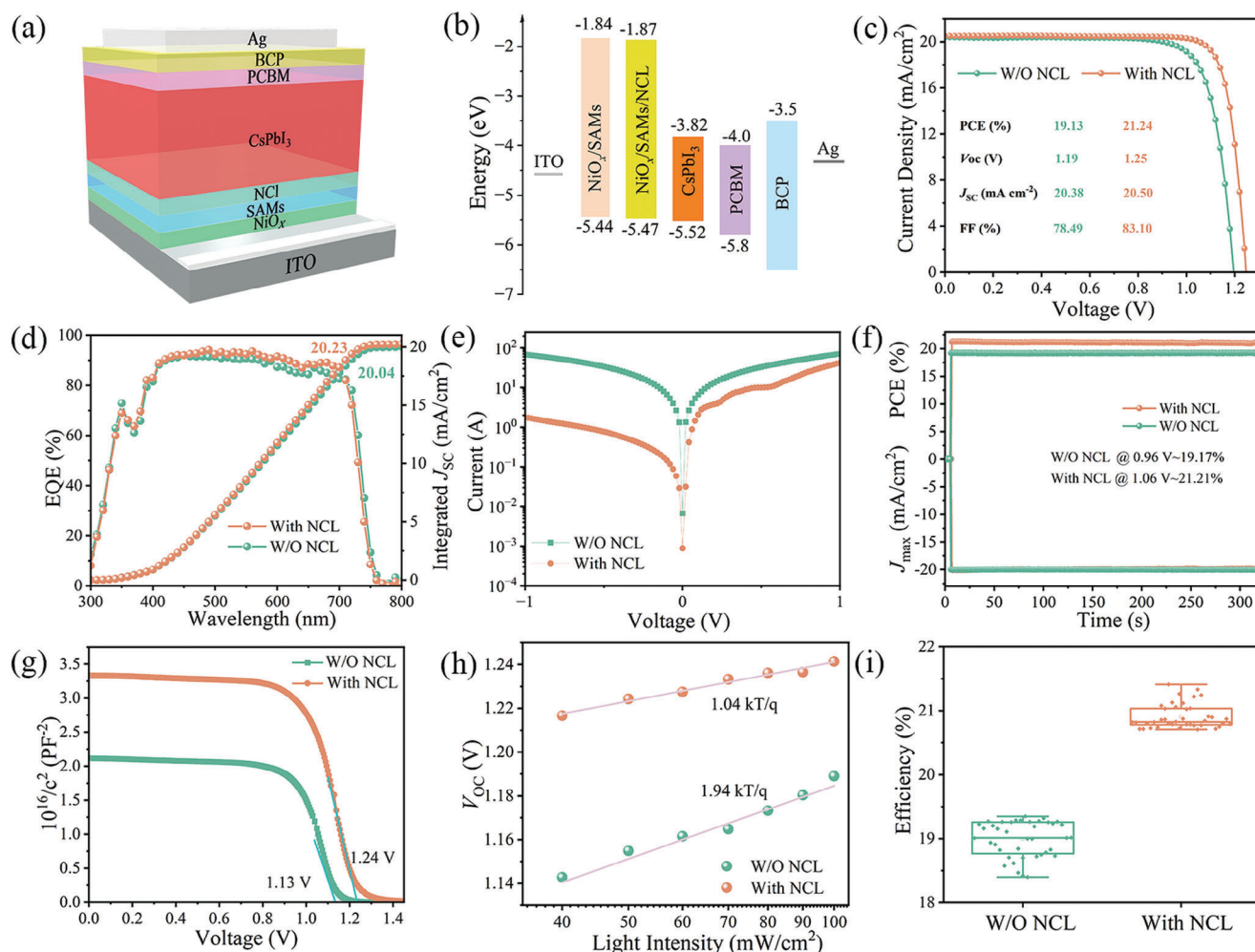


Figure 4. a) Device structure of ITO/NiO_x/SAMs/NCL/CsPbI₃/PCBM/BCP/Ag. b) The schematic energy level arrangement. c) *J*-*V* curves, d) EQE curves, e) the current versus voltage in dark condition, f) the stable output curves, g) the *C*-*V* curves, and h) *V*_{OC} versus light intensity for the inverted CsPbI₃ PSCs without and with NCL modification, respectively. i) The statistic distributions of PCE for 40 inverted CsPbI₃ PSCs without and with NCL modification.

NCL obtained from the external quantum efficiency (EQE) spectra (Figure 4d) are 20.04 and 20.23 mA cm⁻², respectively, consistent with *J*_{sc} values in the *J*-*V* curves. The dark-state *J*-*V* curves of devices are shown in Figure 4e, the reverse current density of the target device was reduced obviously compared with the device without NCL modification, indicating increased shunt resistance and suppressed leakage current in the target device.^[25]

The stable power output (SPO) of device was measured at maximum power point under continuous one sun illumination. As depicted in Figure 4f, the target device outputs a PCE of 21.21% at a bias of 1.06 V, which is consistent with the value obtained from the *J*-*V* curves and higher than that of the control device (19.17%, 0.96 V).^[26] The space-charge limited current (SCLC) method was employed to determine the trap-state densities (*N*_{trap}) of the devices. As shown in Figure S11 (Supporting Information), the *N*_{trap} decreases significantly from 2.67 × 10¹⁶ cm⁻³ for the control device to 1.71 × 10¹⁶ cm⁻³ for the target device, suggesting that NCL can effectively inhibit the defect formation in CsPbI₃ films.^[27]

To validate the impact of NCL treatment on *V*_{OC}, the capacitance-voltage (*C*-*V*) curves of the devices with and without NCL treatment were characterized (Figure 4g), the built-in potentials (*V*_{bi}) were obtained by using the Mott-Schottky equation $C^{-2} = 2(V_{bi} - V)/(A^2 e \epsilon \epsilon_0 N_A)$.^[28] The *V*_{bi} improved from 1.13 V in the control device to 1.24 V in the target device. The enhanced *V*_{bi} (110 mV) should be responsible for the improved *V*_{OC} of the target device, which is primarily attributed to the suppressed defects and proper energy level alignment after NCL modification, indicating a fast charge transport at the buried interface.^[29] The dependence of *V*_{OC} on light intensity was examined to reveal the charge recombination mechanism. As shown in Figure 4h, the ideality factors (*n*) of devices without and with NCL treatment were calculated to be 1.94 and 1.04 *k*_B*T*/*q*, respectively. The *n* value of the target device is closer to 1 *k*_B*T*/*q*, demonstrating a reduced trap-assisted recombination.^[30] Additionally, by counting 40 independent devices for each type of inverted PSCs, the target devices showed higher average PCEs of 20.94% than 19.01% of control devices (Figure 4i), indicating the excellent

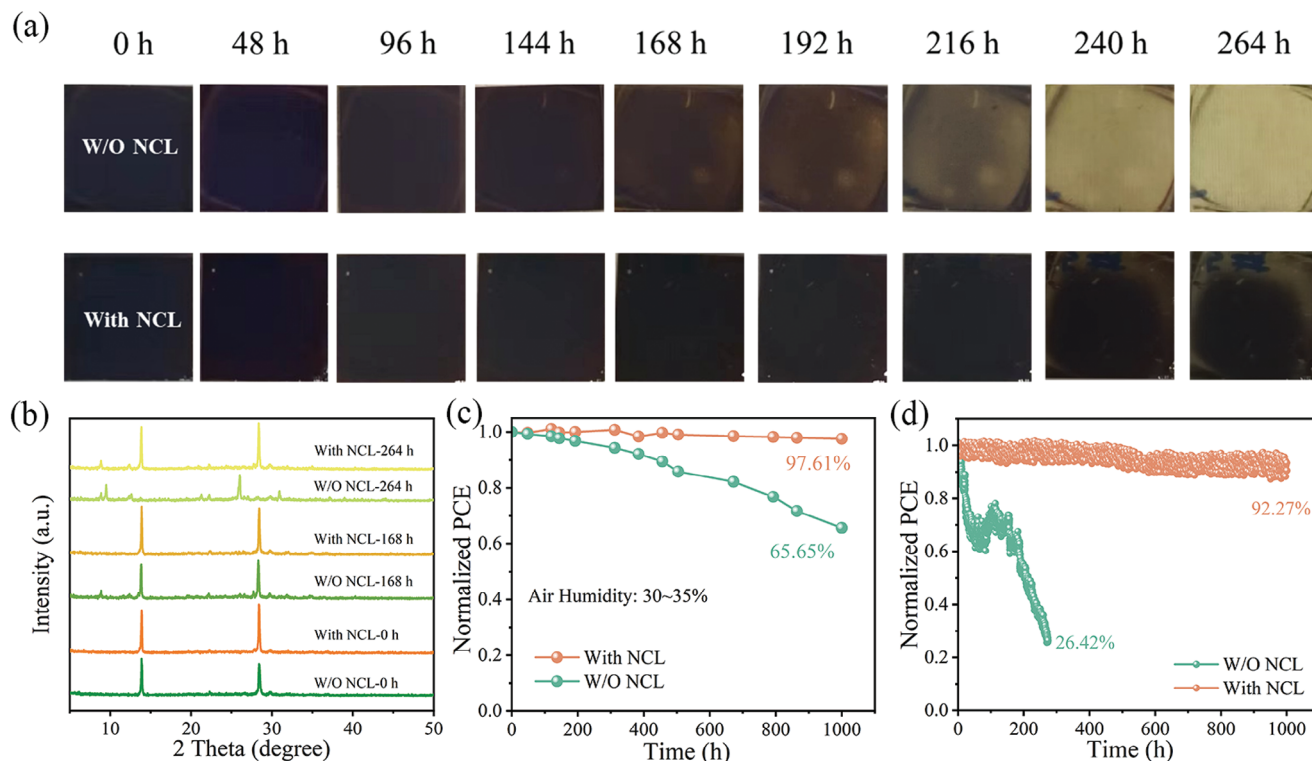


Figure 5. a) The photographs for CsPbI₃ films without and with NCL modification aging for different times at 30–35% RH. b) XRD patterns of CsPbI₃ films without and with NCL treatment at 35% RH for different times. c) The air stability of unencapsulated PSCs without and with NCL modification. d) The MPP tracking curves of encapsulated PSCs without and with NCL modification under continuous light illumination (100 mW cm⁻²) with 30–35% RH for 1000 h.

reproducibility. The statistics of V_{OC} , FF, J_{SC} and PCE of devices with different concentrations of NCL treatment are shown in Figure S12 (Supporting Information).

Apart from the device efficiency, long term stability is another crucial factor for inverted PSCs. To investigate the effect of NCL treatment on the stability of CsPbI₃ inverted PSCs, tests were performed on both perovskite films and devices. Figure 5a displays photographs of the control and NCL-treated CsPbI₃ films stored in air for 264 h with a relative humidity (RH) of 30% and a temperature of 25 °C. The NCL-treated film exhibits better air stability and maintains the black phase throughout the entire period. In contrast, the film without NCL treatment begins to decompose at 168 h, and completely turns into the δ -phase after 264 h. Furthermore, XRD measurement was conducted to illustrate the film stability more accurately. As shown in Figure 5b, both the control and optimized films show prominent peaks initially. However, the control film shows a δ -phase peak after 168 h and is completely devoid of α -phase peaks after 264 h, which is consistent with the results observed in Figure 5a. In term of device stability, the control device witnessed a drop to 65.65% initial PCE after undergoing 1000 h aging in air under ~35% RH and 25 °C, while the target PSC maintained 97.61% initial PCE under the same condition (Figure 5c). The impact of NCL modification on the light stability of devices was also investigated. As shown in Figure 5d, the encapsulated devices were subjected to continuous light exposure (100 mW cm⁻²) under maximum power point (MPP) tracking. The target device maintained 92.27% initial ef-

iciency after 1000 h MPP tracking. In contrast, the efficiency of control device dropped to only 26.42% after 270 h.^[31]

3. Conclusion

In summary, NCL was introduced to improve the performance of inverted CsPbI₃ PSCs via preventing SAMs agglomeration for a homogenous SAMs film with a hydrophilic surface. Thanks to the strong interactions between NCL and SAMs, the NCL treatment not only enables the uniform SAMs distribution, but also conducive to the deposition of perovskite film. In addition, NCL also has a strong interaction with perovskite film, which is facilitated with carrier transfer and nonradiative recombination restraint. As a result, the NCL-treated PSCs exhibit a champion PCE of 21.24%, which is the highest efficiency of inverted CsPbI₃ PSCs reported so far. The optimized PSCs maintain 97.61% initial PCE after 1000 h aging in ambient conditions and 92.27% initial PCE after 1000 h MPP tracking. This work provides a new strategy for improving the PCE and stability of inverted CsPbI₃ PSCs through SAMs optimization.

4. Experimental Section

Materials: Hydrogen lead triiodide (HPbI₃, 99.99%) and Cesium iodide (CsI, 99.99%) were purchased from Xi'an Yuri Solar Co., Ltd. [2-(9H-Carbazol-9-yl)ethyl]phosphonic Acid (2PACz > 98.0%) and

[2-(3,6-Dimethoxy-9H-carbazol-9-yl)ethyl]phosphonic Acid (MeO-2PACz > 98.0%) were purchased from Tokyo Chemical Industry (TCI). Indium-Tin Oxide (ITO) glass, Nickel Oxide (NiO_x), [6,6]-phenyl-C61-butyric acid methyl ester (PC₆₁BM), and bathocuproine (BCP) were purchased from Advanced Election Technology Co., Ltd. Niobium pentachloride (NbCl₅), dimethyl sulfoxide (DMSO, 99.0%), *N,N*-dimethylformamide (DMF, 99.5%) and isopropyl alcohol (IPA, 99.7%) were purchased from Shanghai Aladdin Biochemical Technology Co., Ltd. Chlorobenzene (CB, 99.8%), was purchased from Sinopharm Chemical Reagent Co., Ltd.

Preparation of the Solution of SAMs: 1.00 mmol L⁻¹ 2PACz and 1.00 mmol L⁻¹ MeO-2PACz solutions of IPA were prepared respectively. The solutions were mixed in a 1:1 ratio after stirring for 2 h.

Preparation of NCL: Dissolving the NbCl₅ powder (0.50, 1.00, and 2.00 mg) into 1 mL IPA, and then shaking the solution until NbCl₅ was completely dissolved. The obtained solution was filtered through a 0.45 μm syringe filter for later use.

Preparation of Perovskite Precursor Solution: The perovskite precursor solution (0.745 M) was prepared by mixing CsI, HPbI₃ with a molar ratio of 1:1 in a mixture of DMSO and DMF with a 1:9 ratio by volume. After stirring for 12 h with an oscillator device, the solution was filtered through a 0.45 μm syringe filter in the N₂ glove box.

Device Fabrication: First, wipe the ITO glass clean with a dust-free cloth dipped in ethanol, then place it in an IPA solvent and sonicate for 30 min. Before use, the ITO substrates were dried with a nitrogen gun and then treated with UV ozone (UVO) for 15 min. For the NiO_x nanoparticles (20 mg mL⁻¹ in ultra pure water), 70 μL of the solution was spread over the ITO substrate at 4000 rpm for 30 s and subsequently annealed at 100 °C for 20 min in air (RH = 35%, T = 25 °C). Transfer the annealed NiO_x substrate to an N₂ glove box, spin coat the previously prepared SAM solution at 4000 rpm for 30s, and then anneal at 100 °C for 10 min. For optimized device, spin coat the previously prepared solution of NbCl₅ onto the substrate of NiO_x/SAMs using a 4000 rpm for 30s, and then anneal for 5 min. The perovskite precursor solutions were spin-coated on the ITO/NiO_x/SAMs/ NbCl₅ at 1000 rpm for 10 s and 4000 rpm for 30 s, and then annealed at 160 °C for 50 min in air at 160 °C for 50 min in air at 20–25% RH and T = 20 °C. Transfer the annealed perovskite film back into the N₂ glove box and spin PCBM (20 mg/mL) on top of the annealed perovskite film at 2000 rpm and 30s. Then, spin coat BCP on the top at 4000 rpm for 30 s. Finally, 80 nm Ag was thermal evaporated onto the BCP layer, the active area of the device is determined by a shadow mask of 0.09 cm².

Device Characterization: The SEM images were obtained by a HITACHI, SU-8020 field-emission SEM. Water contact angle measurements were performed on measuring optical contact angle tester (Kruss, Germany). The AFM images were taken by BRUKER Atoms Force Microscope under an air atmosphere. The crystallographic characterizations: XRD patterns were tested by a D/MAX 2400 Diffractometer with Cu Kα radiation (1.5405 Å). The UV–vis absorption spectra were recorded using a UV–vis NIR spectrophotometer (PerkinElmer, Lambda 950). The steady-state PL and TRPL measurement was carried out on Edinburgh Instruments Ltd FLS980 spectrometer with excitation wavelength 510 nm. XPS was taken with ESCALAB250Xi, Thermo Fisher Scientific. The FTIR spectra were measured by PE-Frontier. The *J*–*V* curves were obtained using a Keithley 2400 source under simulated sunlight (AM 1.5G, SAN-ELECTRIC XES-40S2-CE). EQE tests were measured by a QTest Station 2000ADI (Crown-tech, Inc. USA). The distribution map of elements was obtained by X-ray spectroscopy (EDS, Hitachi SU-8020). The space charge limited current (SCLC) was tested on the hole-only devices (FTO/NiO_x/perovskite/Spiro-OMeTAD/Au) by collecting the *I*–*V* characteristics in dark condition. The long-term storing stability of the devices was measured by storing the devices in atmosphere with RH of 30–35%.

Statistics Analysis: All statistical analysis were performed by using Origin 2023, and all data were rounded to two significant digits after the decimal point expected for the DTF calculation results. Data obtained by SEM, AFM, UV-vis, PL, transmittance, contact angle, XPS, *J*–*V* and EQE were the original data without normalization or other transformation. Other data were obtained by transferring their original data according to the calculated equation. The biexponential decay function is performed to TRPL

data by using Origin 2023. Statistics distribution data of device performance including PCE, V_{OC}, J_{SC} and FF were obtained from 40 independent devices.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

Research data are not shared.

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